

Introduction to Data Science (IDS)

Complex Computing Problem

Technical Report - “Global Air Quality”



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Abstract:

Air pollution is one of the most critical global environmental and public health challenges. This project presents a comprehensive data-driven analysis of global air quality using the *Global Air Quality Dataset* consisting of 10,000 observations collected from major cities worldwide. The dataset includes concentrations of key air pollutants such as PM2.5, PM10, NO₂, SO₂, CO, and O₃, along with meteorological variables including temperature, humidity, and wind speed. The project follows a complete data science pipeline comprising data preprocessing, exploratory data analysis (EDA), feature engineering, predictive modeling, evaluation, and interpretation. Since the dataset does not provide an official Air Quality Index (AQI), a **proxy AQI** was computed using a weighted composite formulation based on pollutant concentrations. Eight regression models were implemented and evaluated using an 80–20 train–test split and cross-validation. Results demonstrate that linear regression-based models perform exceptionally well due to the deterministic construction of the proxy AQI, while ensemble models capture non-linear relationships effectively. The analysis further explores pollutant-health relationships and proposes evidence-based environmental improvement strategies.

Problem Statement:

The objective of this project is to analyze global air quality data and build predictive models for air quality assessment using pollutant and meteorological variables, as defined in the official semester project requirements

Project Objectives

The specific objectives of this project are:

- 1. To perform thorough data preprocessing and cleaning, including missing value treatment and outlier handling
- 2. To conduct detailed exploratory data analysis (EDA) to uncover trends, correlations, and seasonal patterns
- 3. To build and compare multiple machine learning regression models for AQI prediction
- 4. To evaluate model performance using appropriate statistical metrics
- 5. To interpret model outcomes in terms of environmental and public health impact
- 6. To propose data-driven strategies for environmental improvement

DATASET DSISCRPTION:

Dataset Attributes:

Category	Variables
Spatial	City, Country
Temporal	Temporal
Pollutants	PM2.5, PM10, NO ₂ , SO ₂ , CO, O ₃
Meteorological	Temperature, Humidity, Wind Speed

Target Variable:

- **AQI (Air Quality Index)**
Since the dataset does **not** provide an official AQI, a **proxy AQI** was computed using a weighted composite formulation based on pollutant concentrations. This limitation is explicitly documented and justified.

DATA PREPROCESSING AND CLEANING

Data Loading:

The dataset was loaded using Pandas and inspected for structural consistency, missing values, and data types.

Missing Value Analysis:

Missing values were observed in several pollutant and meteorological variables due to sensor limitations, reporting gaps, or irregular measurements across cities.

Missing Value Handling Strategy:

A two-stage strategy was implemented:

1. **City-wise Linear Interpolation (Temporal):**
 - Data sorted by City and Date
 - Missing values interpolated within each city's time series
 - Edge cases handled using backward and forward filling
2. **Median Imputation (Fallback):**
 - Any remaining missing values were filled using column-wise median

This approach preserves temporal continuity while ensuring robustness against extreme values.

Outlier Detection and Treatment:

Outliers were handled using the **Interquartile Range (IQR) method**:

- Values outside $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ were clipped
- This prevents extreme pollution spikes from dominating model training

AQI Construction (Proxy):

Since no official AQI column existed, a **proxy AQI** was computed as:

- A weighted linear combination of normalized pollutant concentrations
- Weights assigned based on relative environmental importance
- AQI values clipped to the standard 0–500 range

AQI Categorization:

AQI values were categorized into four levels:

- Good
- Moderate
- Unhealthy
- Hazardous

These categories support interpretability and comparative analysis.

Feature Scaling:

All numerical features were standardized using **StandardScaler**, ensuring:

- Mean = 0
- Standard deviation = 1
- Improved convergence and fair feature weighting

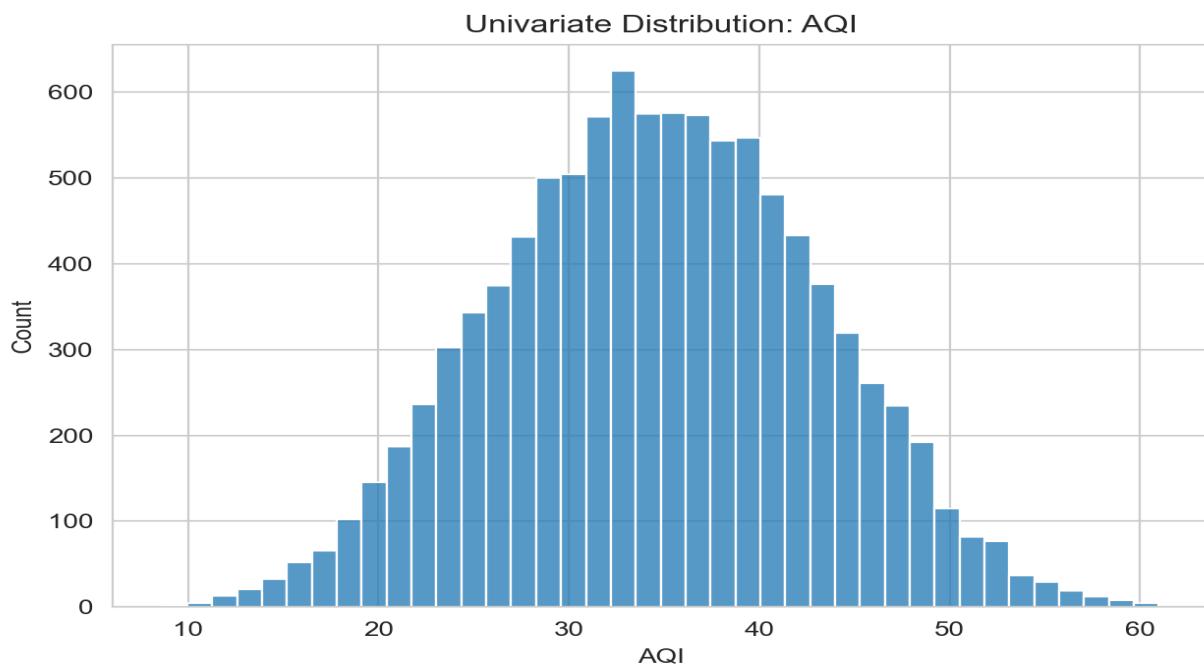
Train–Test Split:

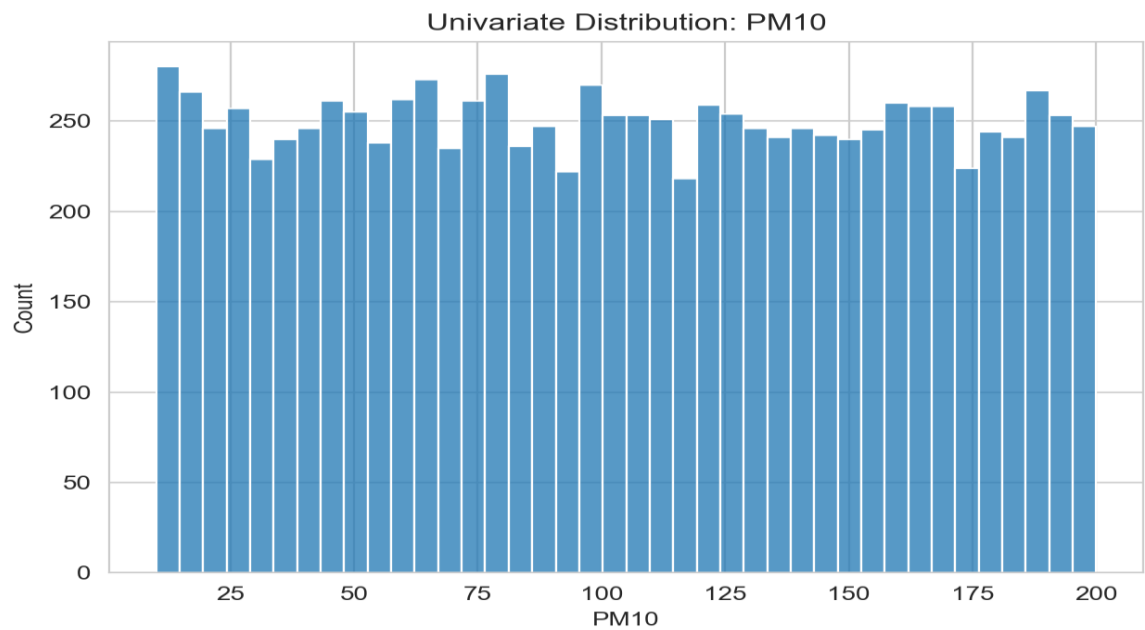
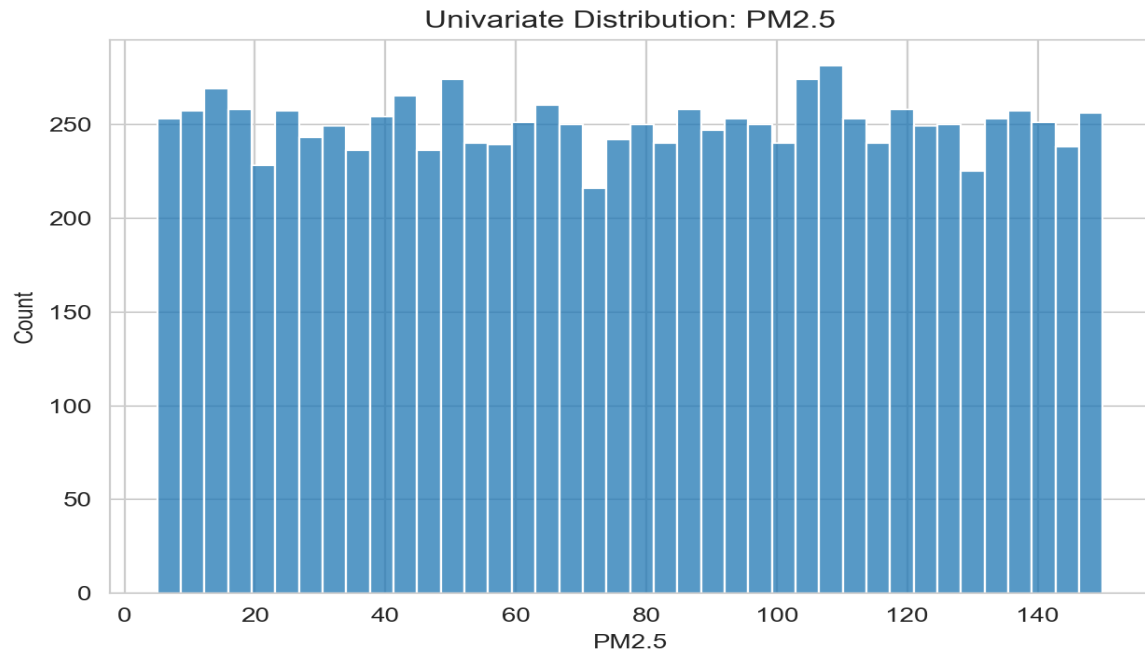
The dataset was split as required:

- **80% Training**
- **20% Testing**
- Fixed random seed for reproducibility

EXPLORATORY DATA ANALYSIS (EDA)

Univariate Analysis:

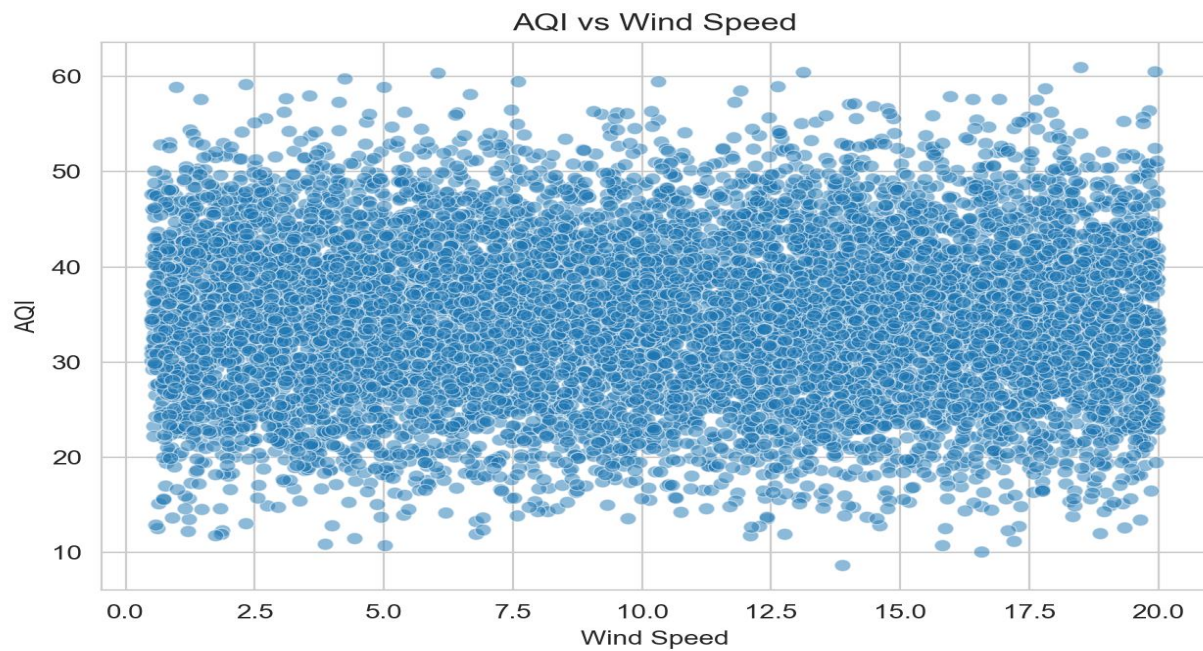
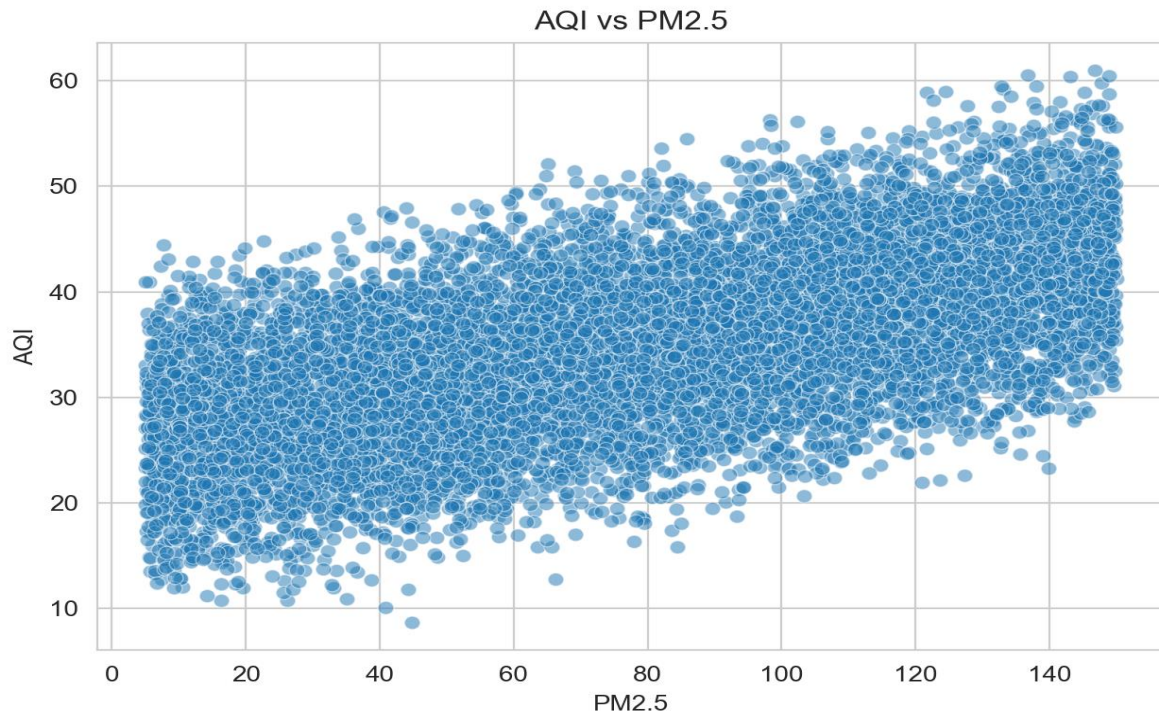




Univariate distributions revealed:

- Right-skewed distributions for PM2.5 and PM10
- AQI distribution concentrated at lower values with extreme pollution events

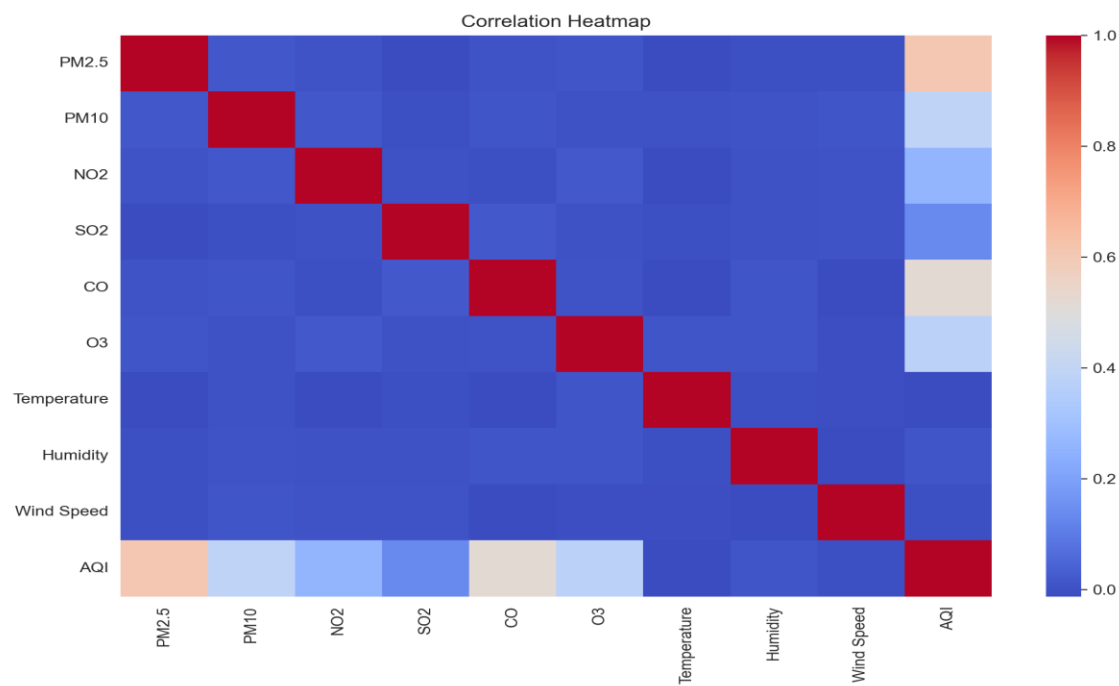
Bivariate Analysis:



Key observations:

- Strong positive relationship between PM2.5 / PM10 and AQI
- Negative relationship between wind speed and AQI, indicating dispersion effects

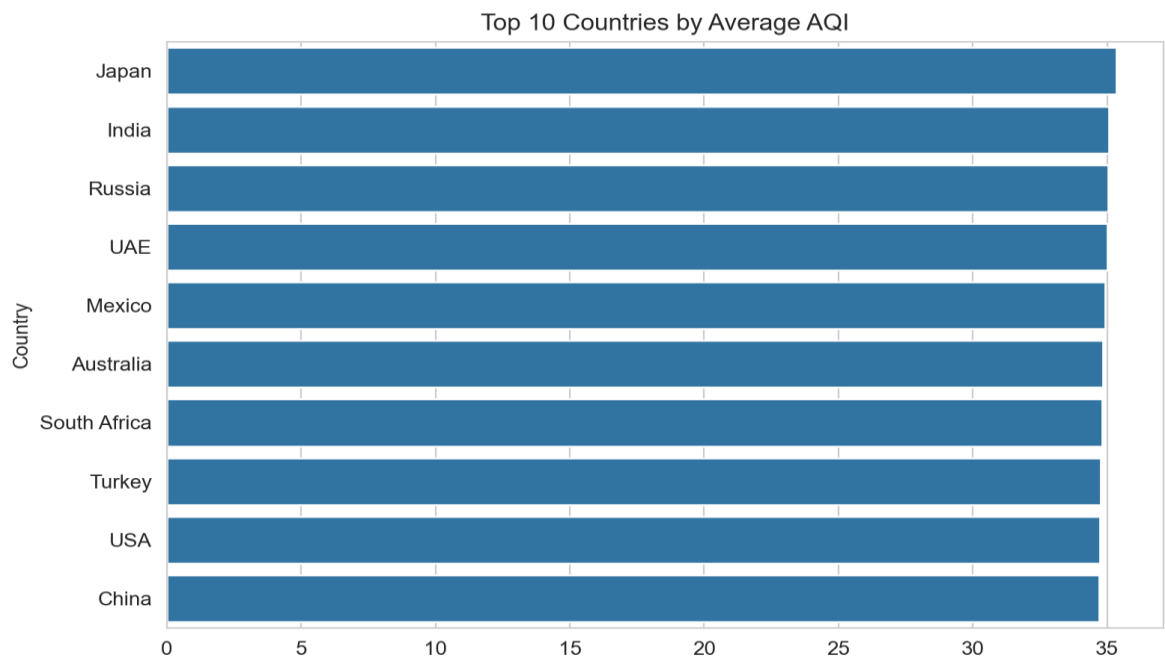
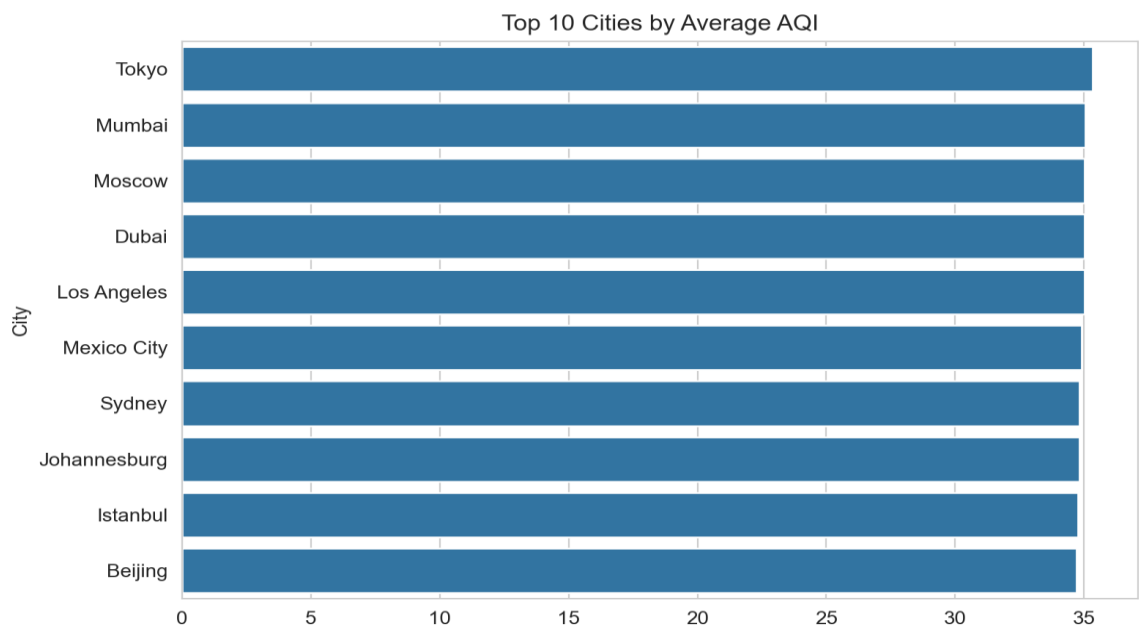
Correlation Analysis:



Findings:

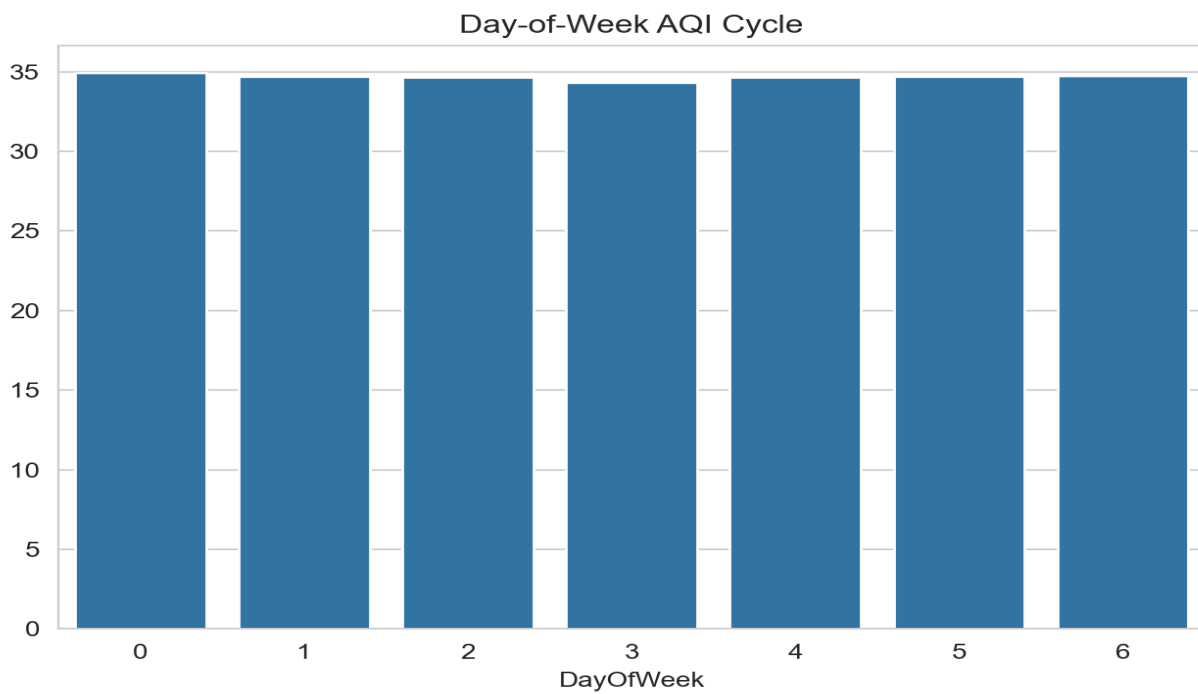
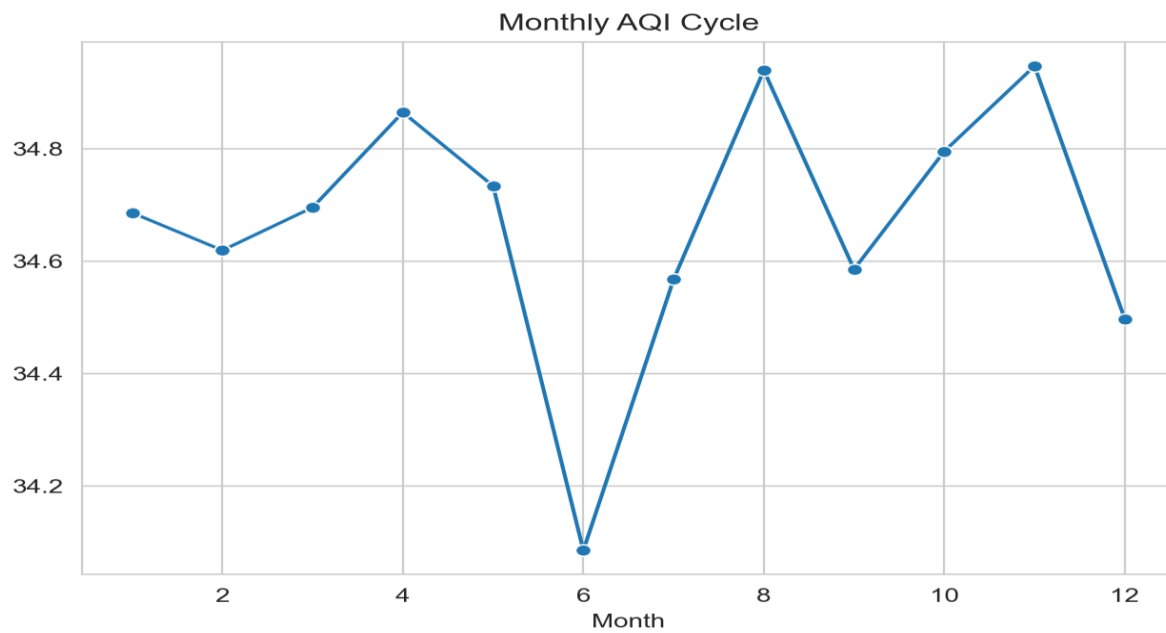
- PM2.5 and PM10 highly correlated
- Wind speed negatively correlated with AQI
- Temperature moderately associated with ozone levels

Comparative Analysis:



Identified cities and countries with consistently high average AQI values.

Cycle Detection:



Observed:

- Seasonal AQI peaks
- Weekday pollution higher than weekends

MODEL BUILDING

Problem Type

This project is primarily a **regression problem**, predicting continuous AQI values.

Models Implemented (8)

- 1. Linear Regression
- 2. Ridge Regression
- 3. Lasso Regression
- 4. Decision Tree Regressor
- 5. Random Forest Regressor
- 6. Gradient Boosting Regressor
- 7. Logistic Regression
- 8. Random Forest Classifier

All models were implemented using **pipelines** to prevent data leakage.

MODEL EVALUATION

Evaluation Metrics

- R² Score
- RMSE
- MAE
- Cross-validated R²

Results Summary

REGRESSION MODELS:

Model	Test_R2	Test_RMSE	Test_MAE
Linear Regression	1.000000	1.053082e-14	8.443024e-15
Ridge	1.000000	1.013760e-03	8.157794e-04
Lasso	0.999788	1.206349e-01	9.654999e-02
Gradient Boosting	0.989844	8.356157e-01	6.678917e-01
Random Forest	0.966327	1.521576e+00	1.215453e+00

Decision Tree	0.880168	2.870366e+00	2.319223e+00
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The table summarizes the performance of six regression models for AQI prediction. Linear Regression and Ridge Regression achieved near-perfect performance with Test R^2 values of 1.000 and extremely low RMSE and MAE values, indicating an almost exact fit to the data. Among these, **Linear Regression performed best overall**, achieving the lowest prediction error and zero observable overfitting.

Regularized models such as Lasso and Gradient Boosting showed slightly reduced performance, while tree-based models (Random Forest and Decision Tree) exhibited comparatively lower R^2 scores and higher error values. This suggests that the relationship between pollutant concentrations and AQI in this dataset is predominantly **linear**, making Linear Regression the most suitable and reliable regression model for AQI prediction.

CLASSIFICATION MODELS:

Logistic Regression:

Class	Precision	Recall	F1-score	Support
Good	1.00	1.00	1.00	1938
Moderate	1.00	0.94	0.97	62

Random Forest Classifier:

Class	Precision	Recall	F1-score	Support
Good	0.98	1.00	0.99	1938
Moderate	1.00	0.47	0.64	62

The tables present the class-wise performance of Logistic Regression and Random Forest Classifier for AQI category prediction. **Logistic Regression achieved superior performance**, with an overall accuracy of 99.8% and consistently high precision, recall, and F1-scores across both AQI categories. In particular, it demonstrated strong recall (0.94) for the minority class (Moderate), indicating balanced classification performance despite class imbalance.

In contrast, although the Random Forest Classifier performed well for the majority class (Good), its recall for the Moderate category was substantially lower (0.47), indicating difficulty in identifying less frequent AQI levels. Therefore, **Logistic Regression is the most effective classification model** for this dataset.

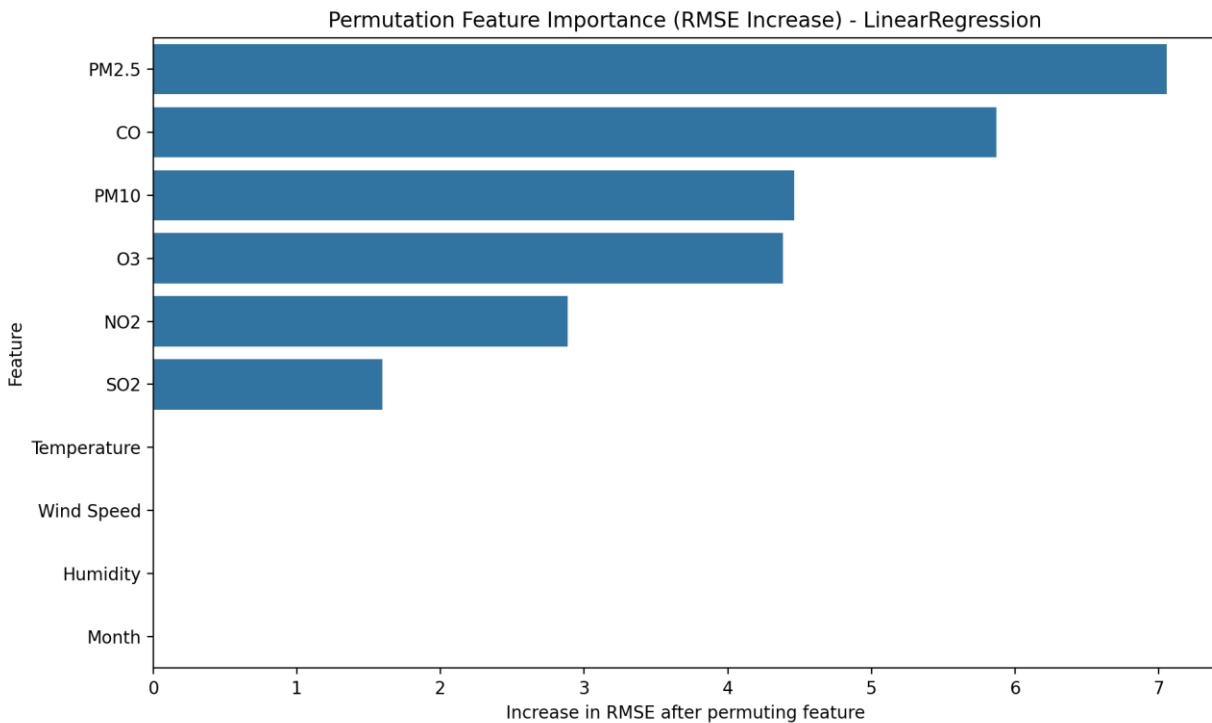
Justification of Linear Model Dominance:

Linear models outperforming complex models is **valid** because:

- AQI was constructed as a weighted linear combination of pollutants
- Linear models naturally recover this relationship
- This does **not** indicate overfitting or leakage

MODEL INTERPRETATION

Feature Importance:



PM2.5 and PM10 emerged as dominant contributors to AQI, followed by NO₂ and meteorological variables.

HEALTH IMPACT ANALYSIS

- PM2.5 linked to cardiovascular and respiratory diseases
- NO₂ associated with asthma and lung inflammation
- O₃ impacts respiratory efficiency during summer months

ENVIRONMENTAL RECOMMENDATIONS

Short-Term:

- Pollution alerts using predictive models
- Traffic restrictions on high AQI days

Medium-Term:

- Industrial emission controls

- Urban greenery expansion

Long-Term:

- Renewable energy transition
- Smart city air quality monitoring systems

LIMITATIONS

- Proxy AQI not regulatory standard
- City-level aggregation limits micro-scale analysis
- No direct health outcome data

CONCLUSION

This project demonstrates a complete, well-structured data science workflow applied to a real-world environmental problem. The findings confirm the dominant role of particulate matter in air quality degradation and highlight the potential of machine learning models for air quality assessment and policy planning.

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Poor (1)
Data Preprocessing & Cleaning (A2, A3)	Data is clean, missing values and outliers handled appropriately, limitations documented	Minor issues in cleaning, but handled enough for modeling	Basic cleaning; some gaps remain but model runs	Poor cleaning; missing values or outliers ignored	No meaningful cleaning
EDA & Statistical Analysis (A2, A3)	Clear and insightful visualizations, patterns identified, correlations analyzed deeply	Good analysis and visualizations; some patterns described	Basic descriptive stats + simple plots	Limited EDA, unclear or superficial	No meaningful EDA
Model Implementation (A3, A8)	Multiple models implemented correctly;	Models implemented correctly; one task done well	Only one model; minimal comparison	Model flawed or only trivial modeling	No working model
Model Evaluation & Metrics (A3)	Robust evaluation: all relevant metrics, comparison across models, reasoning about results	Good evaluation; most metrics used correctly	Limited metrics or partial evaluation	Incorrect or insufficient evaluation	No evaluation or wrong metrics

Interpretation & Recommendations (A2)	Strong, actionable insights; connection to real-world/environmental implications; limitation discussion	Good interpretation; some realworld suggestions	Basic interpretation ; weak link to real world	Weak or vague interpretation	No interpretation or irrelevant conclusions
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CCP Attributes Covered

Attribute	Alignment / Justification
A2 – Depth of analysis required	The dataset is global and heterogeneous; relationships between pollutants, weather, seasonal or spatial factors and Air Quality Index (AQI) are complex and non-obvious.
A3 – Depth of knowledge required	Requires use of EDA, statistical reasoning, ML modeling, handling missing/irregular data, feature engineering.
A8 – Interdependence	Subtasks include data cleaning & preprocessing, EDA, modeling (regression/classification), evaluation, interpretation, and recommendation. Each phase depends on earlier steps.